FINC5001
Module 3

Presented by
Craig Mellare
University of Sydney Business School

THE UNIVERSITY OF SYDNEY
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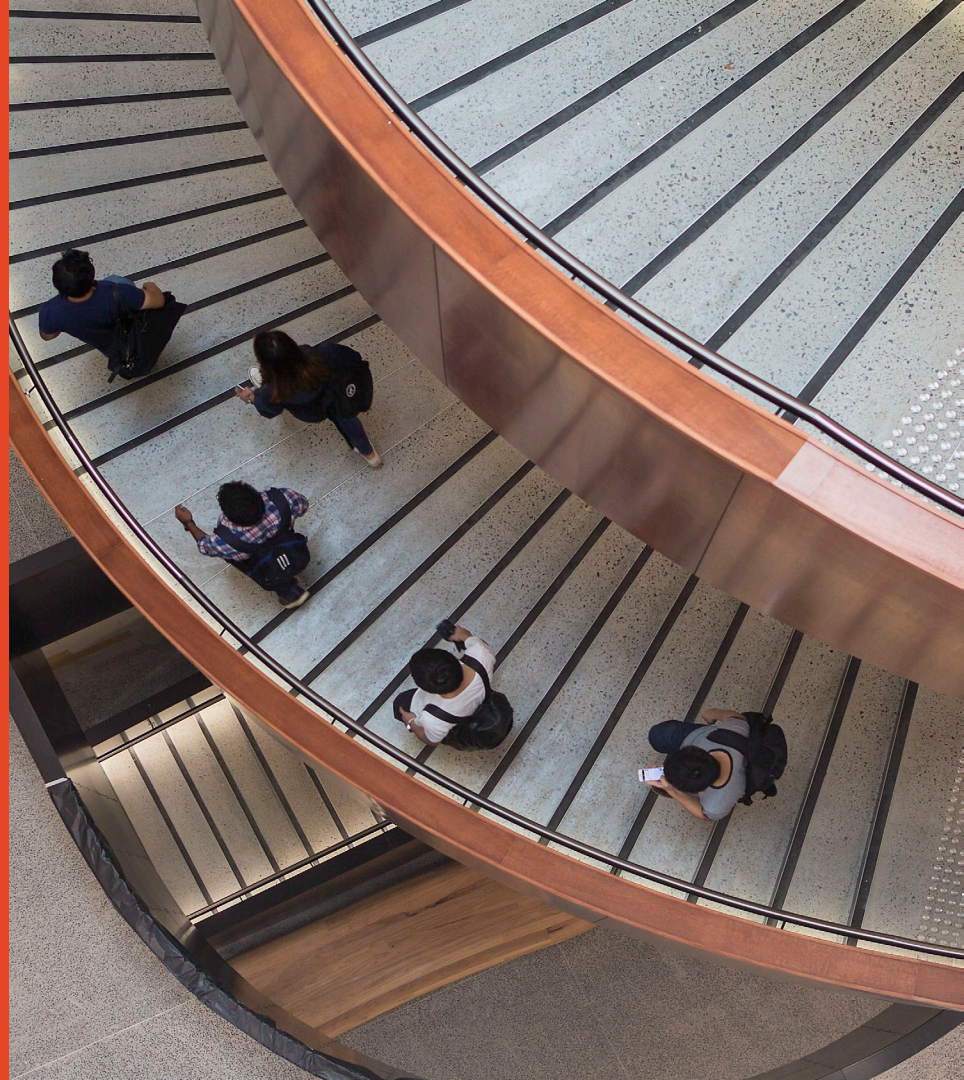


This Lecture

- By the end of this lecture, you should:
  - Be able to determine the present value of an ordinary annuity, annuity due and deferred annuity
  - Be able to determine the future value of an ordinary annuity, annuity due and deferred annuity
  - Be able to calculate equivalent annuities, understand their use and understand the assumption behind the technique
  - Be able to determine the present value of a perpetuity and a growing perpetuity

Annuities

- Annuities are a special case of multiple cash flow streams, where the cash flows are of equal size and occur at regular time intervals
- Examples include:
  - A constant superannuation payment made to retirees on a monthly basis throughout the remainder of their lives
  - Rentals paid in arrears
  - Interest payments on a bond

Annuities (cont)

Asset 1 : Cash flows received annually

Year	0	1	2	3	4	5	6	7	8
			$2.3	$2.3	$2.3	$2.3	$2.3	$2.3	

Asset 2 : Cash flows received every two years

Year	0	1	2	3	4	5	6	7	8
			$4		$4		$4		$4

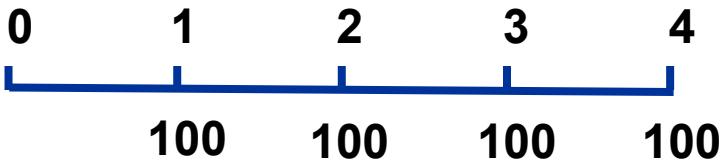
Both of these cash flows are annuities

Annuities (cont)

- Types of annuity:
  - Ordinary annuity
    - Cash flows occur at the end of each period and, hence, the first cash flow occurs at the end of the first year
  - Annuity due
    - Cash flows occur at the beginning of each period and, hence, the first cash flow occurs immediately
  - Deferred annuity
    - The first cash flow in an annuity is delayed by x periods

Future value of an annuity

- A four-year ordinary annuity paying $100 cash flows would look like this



- The future value of the four payments of this annuity can be calculated by compounding each cash flow forward to the fourth year and adding them together

Future value of an annuity (cont)

Example

If you receive $100 at the end of each year for four years and reinvest the cash flows at 5% p.a. compounded annually, what is the future or accumulated value?

FV_1	$\$100 \times (1 + r)^3$	$\$100 \times 1.1576$	$115.76
FV_2	$\$100 \times (1 + r)^2$	$\$100 \times 1.1025$	$110.25
FV_3	$\$100 \times (1 + r)^1$	$\$100 \times 1.05$	$105.00
FV_4	$100	$100	$100.00
		Total	$431.01

Future value of an annuity

– To determine the accumulated value of an ordinary annuity of $PMT :

$$FV = PMT\left[\frac{(1+i)^n - 1}{i}\right]$$

where:

FV = the accumulated or future value of the annuity

PMT = the cash flow received/paid under the annuity

n = the number of cash flows that form the annuity

i = the compound interest rate per period

Future value of an annuity (cont)

Example

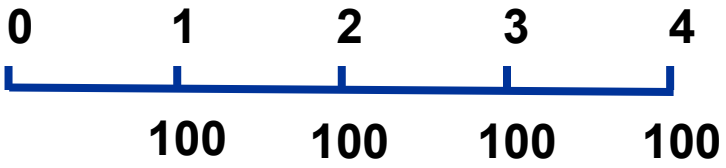
At the end of each year, you place $500 in an account that earns 5% interest p.a. compounded annually. How much will be in the account at the end of five years?

$$FV = PMT\left[\frac{(1+i)^n - 1}{i}\right]$$

$$= 500\left[\frac{(1+.05)^5 - 1}{.05}\right] = \$2763$$

Present value of an annuity

- The present value of an annuity can also be found by discounting the individual cash flows



- The present values of the four components of this annuity can be calculated by discounting each cash flow to the present and adding them together

Present value of an annuity (cont)

Example

If you receive $100 at the end of each year for four years, what is the value of these cash flows if the current rate of interest is 5% p.a. compounded annually?

PV_1	$\$100 \times (1 + r)^{-1}$	$100 / 1.05	$95.24
PV_2	$\$100 \times (1 + r)^{-2}$	$100 / 1.1025	$90.70
PV_3	$\$100 \times (1 + r)^{-3}$	$100 / 1.1576	$86.39
PV_4	$\$100 \times (1 + r)^{-4}$	$100 / 1.2155	$82.27
		Total	$354.60

Present value of an annuity (cont)

– Alternatively, to determine the present value of an ordinary annuity of *$PMT*:

$$PV = PMT\left[\frac{1-(1+i)^{-n}}{i}\right]$$

where:

PV = the present value of the annuity

PMT = the cash flow received/paid under the annuity

n = the number of cash flows that form the annuity

i = the compound interest rate per period

Present value of an annuity (cont)

Example

What is the present value of a security that pays you $0.40 per half year for 15 years, if a competing investment is offering 4.04% p.a. compounded annually?

- 4.04% p.a. = 2% per half year [Effective Rate]

$$PV = \$0.40\left[\frac{1-(1+0.02)^{-30}}{0.02}\right] = \$8.96$$

Annuity due

- To date we have assumed that payments occur at the end of each period throughout the annuity
  - *ordinary annuity*
- Often payments are made at the beginning of each period (e.g. rental agreements)
  - *annuity due*
- To determine the present value of an annuity due:

$$PV = PMT + PMT\left[\frac{1-(1+i)^{-(n-1)}}{i}\right]$$

Present value of an annuity due

Example

What is the present value of a $1000 annuity that pays five regular payments when the interest rate is 10% p.a. with the first payment due immediately?

This is equivalent to an amount of $1000 now plus an ordinary annuity of four payments of $1000

$$PV = 1000 + 1000\left[\frac{1-(1+.10)^{-(5-1)}}{.10}\right] = \$4170$$

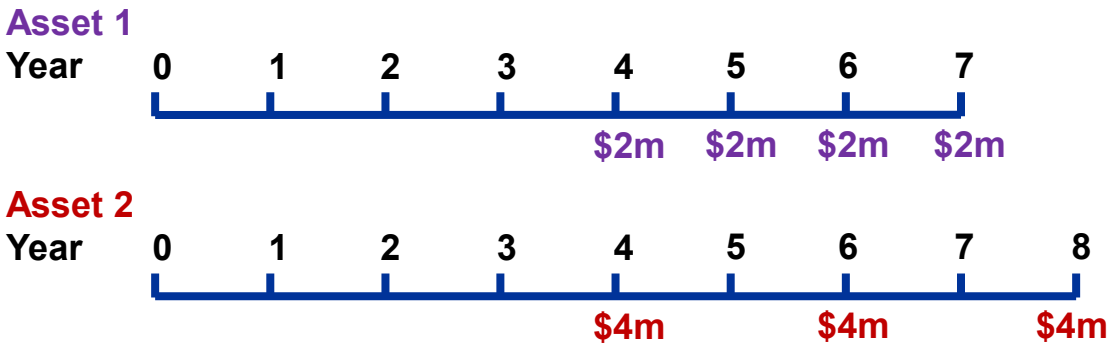
Ordinary Annuity Vs Annuity Due

Year	0	1	2	3	4	5	6	7
		$2k	$2k	$2k	$2k	$2k	$2k	$2k

Year	0	1	2	3	4	5	6	7
	$2k	$2k	$2k	$2k	$2k	$2k	$2k	

Deferred annuities

- A deferred annuity is an ordinary annuity that does not begin in one period's time, but at a later date



  - Asset 1 is an ordinary annuity that begins in three years (first cash flow is in four years), while Asset 2 is an ordinary annuity that begins in two years

Deferred annuities (cont)

- To determine the present value of a deferred annuity (where *x* is the number of periods before the first cash flow occurs):

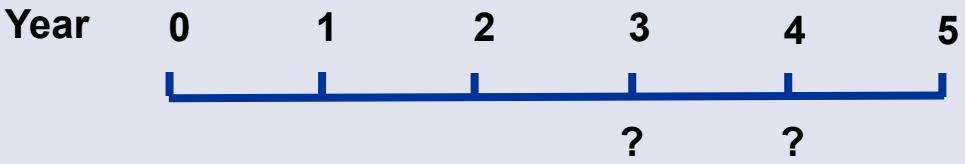
$$PV = \left[PMT \times \frac{1-(1+i)^{-n}}{i} \right] \Big/ (1+i)^{x-1}$$

  - The term in large brackets is the present value of an ordinary annuity of *n* payments
  - This second term gives the present value of the single lump sum calculated from the first term, discounted back *x* - 1 periods

Deferred annuities (cont)

Example

What is the present value of an asset that commences paying cash flows of $2 million in three years' time for three years, when the interest rate is 5%?

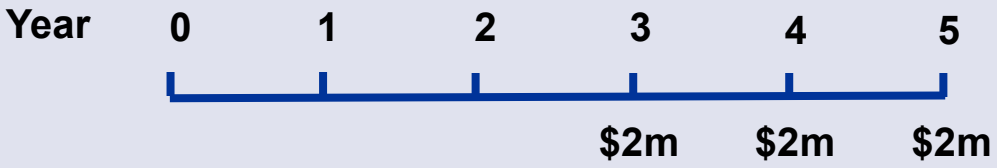


Where is the first cash flow?

Deferred annuities (cont)

Example

What is the present value of an asset that commences paying cash flows of $2 million in three years' time for four years, when the interest rate is 5%?



An ordinary annuity would occur in two year's time, so the above series of cash flows is the equivalent of an ordinary annuity deferred for two years.

Deferred annuities (cont)

Example

What is the present value of an asset that commences paying cash flows of $2 million in three years' time for three years, when the interest rate is 5%?

Step 1: Present value of $2 million payments at end of year 2

$$PV_2 = \$2m\left[\frac{1-(1+.05)^{-3}}{.05}\right] = \$5.45m$$

Step 2: Calculate the present value of this lump sum cash flow occurring at the end of period 2 back to time 0

$$PV_0 = \frac{FV}{(1+i)^n} = \frac{5.45}{(1.05)^2} = \$4.94m$$

Equivalent annuity

- It is sometimes necessary to compare cash flow streams in which cash flows occur at different time intervals or over different lives
- This can be done by converting one or more cash flows to an *equivalent annuity*
- This is done by:
  - Calculating the present value of a cash flow stream
  - Determining an annuity whose present value is equal to this cash flow stream

Equivalent annuities (cont)

- You are looking at buying a new printing machine for your factory. You have two machines to choose from. Both perform the same function. Machine A lasts for 4 years whilst Machine B lasts for 7 years
- The required rate of return is 10% pa
- The costs of running each machine are given on the next slide

Equivalent annuities (cont)

Example (cont)

Machine A

Year	0	1	2	3	4
	-$10k	-$3k	-$3k	-$3k	-$3k

Machine B

Year	0	1	2	3	4	5	6	7
	-$6k	-$4k	-$4k	-$4k	-$4k	-$4k	-$4k	-$4k

Which should we purchase?

Equivalent annuities (cont)

Example (cont)

If we calculate the PV of each machine we get:

Machine A

$$PV_A = -\$10{,}000 - \frac{\$3{,}000}{(1+0.10)^1} - \frac{\$3{,}000}{(1+0.10)^2} - \frac{\$3{,}000}{(1+0.10)^3} - \frac{\$3{,}000}{(1+0.10)^4} = -\$19{,}510$$

Machine B

$$PV_B = -\$6{,}000 - \frac{\$4{,}000}{(1+0.10)^1} - \frac{\$4{,}000}{(1+0.10)^2} - \frac{\$4{,}000}{(1+0.10)^3} - \frac{\$4{,}000}{(1+0.10)^4} - \frac{\$4{,}000}{(1+0.10)^5} - \frac{\$4{,}000}{(1+0.10)^6} - \frac{\$4{,}000}{(1+0.10)^7} = -\$25{,}473$$

Therefore, we might choose Machine A since it has the lowest cost

Equivalent annuities (cont)

Example (cont)

However, this ignores the fact the machines have different lives. After all Machine A does not last as long as machine B. So, we must once convert each cash flow stream in an equivalent annuity

Equivalent annuities (cont)

Example (cont)

Machine A

- $PV = PMT([1-(1+i))^{-t}]/i)$
- $\$19,510 = PMT(([1-(1+0.10)^{-4}]/0.10)$
- $PMT = \$6,154.83$

Machine B

- $\$25,473 = PMT(([1-(1+0.10)^{-7}]/0.10)$
- $PMT = \$5,232.29$

Hence Machine B is actually cheaper to run each year than Machine A.

Present value of a growing annuity

– To determine the present value of a growing ordinary annuity of *$PMT*:

$$PV = PMT \times \frac{1}{i-g}\left[1 - \frac{(1+g)^n}{(1+i)^n}\right]$$

where:

PV = the present value of the annuity

PMT = the cash flow received/paid under the annuity

n = the number of cash flows that form the annuity

i = the compound interest rate per period

g = the constant growth rate over the life of the annuity

Present value of a growing annuity (cont)

Example

What is the present value of an annuity that generates an annual cash flow of $5,000 at the end of the year, has a maturity of 3 years, is growing at 6% pa and has a required return of 10% pa compounded annually

$$PV = \$5000 \times \frac{1}{0.10-0.06}\left[1 - \frac{(1+0.06)^3}{(1+0.10)^3}\right] = \$13,147$$

Perpetuities

- Perpetuities are a special type of annuity where the cash flows are expected to go on forever
- The present value of a perpetuity:

$$PV = \frac{PMT}{i}$$

where:

PV = the present value of the perpetuity

PMT = the cash flow received/paid under the perpetuity

i = the compound interest rate per period

Perpetuities (cont)

Example

An investment that pays a cash flow of $1000 every year forever (starting in one year's time) has a required rate of return of 10% pa. What is the present value of the investment?

$$PV = \frac{PMT}{i}$$

$$PV = \frac{\$1000}{0.1} = \$10{,}000$$

Constant Growth Perpetuity

- Constant growth perpetuities are a special type of perpetuity where the cash flows are expected to grow at a constant rate forever
- The present value of a constant growth perpetuity:

$$PV_t = \frac{PMT_{t+1}}{i - g}$$

- Where

  g = the constant growth rate

Constant Growth Perpetuities (cont)

Example

What is the present value of $1billion paid at the end of the year that will grow at a constant rate in perpetuity, assuming a required rate of return of 10% p.a. and a constant growth rate of 4% p.a.?

$$PV_0 = \frac{\$1b}{0.10 - 0.04} = \$16.667b$$

The End

Next week we look at how we apply financial mathematics to the valuation of debt and equity